Host Utilities

TATA

TATA POWER

TATA POWER-DDL
Towards a Greener Tomorrow

BSES
BSES Rajdhani Power Limited

BSES
BSES Yamuna Power Limited

ORGANIZER

ISGF
India Smart Grid Forum

India
SMART UTILITY
Week 2026
10 – 14 March 2026, New Delhi

Supporting Ministries

सत्यमेव जयते
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सत्यमेव जयते
MINISTRY OF POWER
GOVERNMENT OF INDIA
CENTRAL ELECTRICITY AUTHORITY

Session : Digitalization Road Map in Utilities

Presented By

Sandip Pal, Head Distribution Services, TPNODL

Formation of TPNODL

ODISHA ELECTRICITY REGULATORY COMMISSION
BIDYUT NIYAMAK BHAWAN PLOT NO. 4, CHUNUKOLI, SHAILASHREE VIHAR, BHUBANESWAR-751021

****

Present: Shri U. N. Behera, Chairperson
Shri S. K. Parhi, Member
Shri G. Mohapatra, Member

Case No. 9/2021

IN THE MATTER OF: Suo Motu Proceeding on sale of utility of NESCO under Section 20 of the Electricity Act 2003 and for vesting of Utility (NESCO) to the intending purchaser under Section 21 of the said Act.

And

IN THE MATTER OF:

Director (Regulatory Affairs), OERC
…… Designated Petitioner

Vrs.

Principal Secretary to Government,
Department of Energy, Government of Odisha,
Bhubaneswar -1 and Others …..... Respondents

NESCO Power Engineer's Association
…..... Intervenor

ORDER

Date of order: 25.03.2021

1. North Eastern Electricity Supply Company of Odisha Limited (the "NESCO") was incorporated on 19th November 1997 under the Companies Act, 1956. Pursuant to the Odisha Electricity Reforms Act 1995 and Odisha Electricity Reforms Rules 1998, all the assets of GRIDCO pertaining to the distribution business in the Northern Zone of GRIDCO comprising districts of Balasore, Mayurbhanj, Keonjhar, Jajpur, and Bhadrak were transferred to NESCO.

2. On 1st April 1999, 51% (fifty one percent) shares of GRIDCO in NESCO were transferred to BSES Limited selected through competitive bidding process. NESCO continued to be managed by BSES Limited and later by its successor R-Infra

49 %

51 %

TPNODL formed on 1st of April-2021

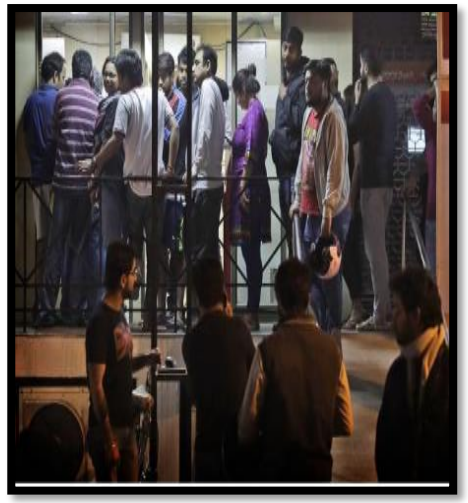
Challenges During Inception

Operational Limitations

- No dedicated laptops/desktops for employees
- One Desktop in 1 dept.
- G-mail usage
- Data storage in Excel sheet (desktop)

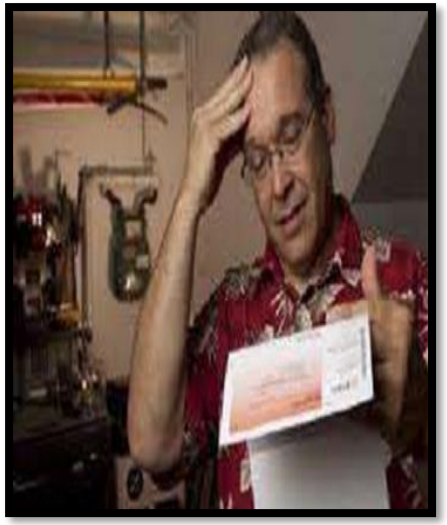
Customer Experience

- No dedicated customer care/call center
- Long queues at offices
- Manual, time-consuming service processes
- High resolution time due to fragmented systems



Billing & Payments

- No digital payment options
- High customer gathering in cash counters
- Lengthy bill correction & tracking Process



Service Challenges

- Overloaded Foot Fall at offices
- No real time update reaching to consumers
- Inefficient complaint handling
- No unified digital platform

"These challenges highlighted the need for a complete digital transformation."

Key Digitalization Initiatives

Operations

- **SAP** Implementation
- **GIS** implemented across all divisions
- AI-driven safety audit using drones
- Online estimate framing, Quality Check, Inhouse Inspection & internal workflows (**SAP/Fiori)**
- IoT-based device for RMU & Autorecloser operational alert to filed operation
- **M-APP** feeder maintenance
- Cyme Dist software for network engineering
- FCC app for no power supply complaint management

Smart Billing & Payments

- Smart Meter Installation – 4.48 Lakhs
- Prepaid metering for Govt. consumers
- 98% **OCR-based meter** reading
- Time-of-Day & e-bill Automated bill revision (Samadhan app)
- QR Code/App/website/UPI digital payment avenues
- My Tata Power app, missed call services

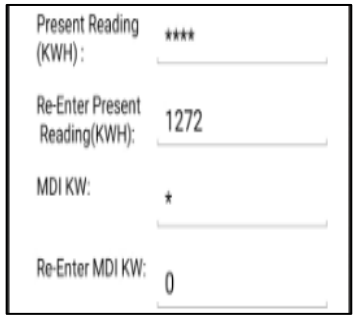
Digital Customer Experience

- **WhatsApp + Chatbot** for instant response
- Online complaint registration & proactive outage alerts
- 100% online new connection journey
- Customer Interaction/compliant handling/feedback through social media desk
- **App-based** meter installation & site survey
- My Tata Power interactive services

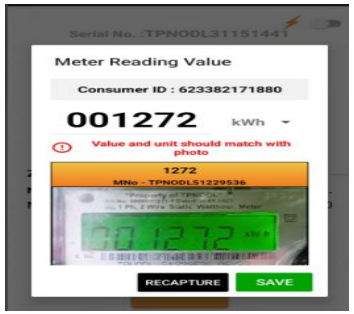
Data & Automation

- **SCADA** for centralized remote operations of substations
- Automated MIS dashboards (billing, loss, collections)
- Daily performance dashboards
- Suraksha Kawach App (SK02) for PTW issuance
- **DWS** – Digital Workplace Switch for secure letter repository, approvals & workflow digitization
- **ADMS** - 1 division (out of 16)

OCR Based Meter Reading



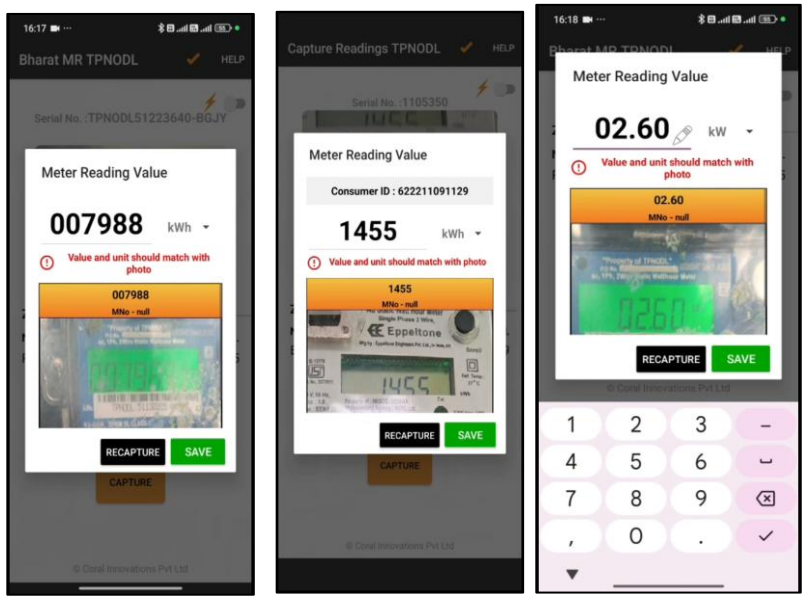
Before:
Manual meter reading → prone to errors, delays, manipulation, and re-visits



After:
AI-Powered **Optical Character Recognition** (OCR) Meter Reading

AI-POWERED OCR FOR METER READING

- Automated extraction of readings from Meter Display
- Eliminates manual entry errors and ensures real-time data validation
- Work even under low-light or unclear display conditions using ML Enhancement

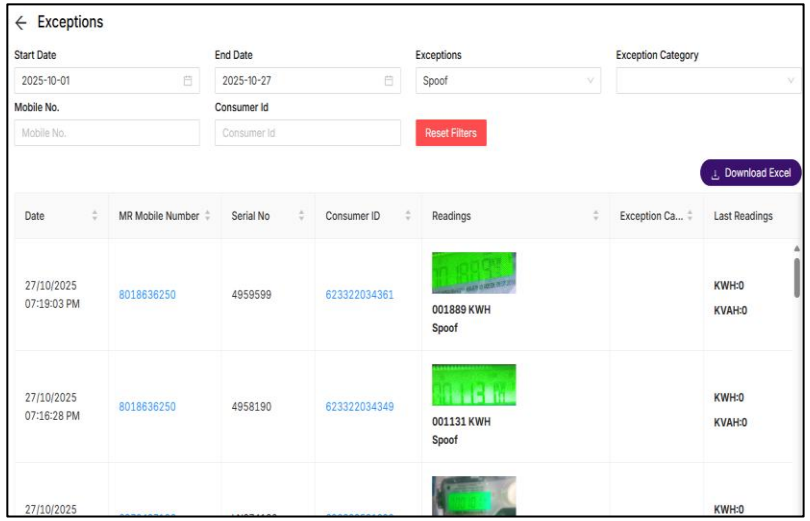


SPOOF &TAMPER DETECTION

- AI Algorithms detects duplicate images, display manipulation, other than meter display , Kwh & Kw Validation
- Alert generated automatically in case of wrong entry

OPERATIONAL EFFICIENCY

- 100% elimination of manual data entry at field level
- Reduced turnaround time for billing & exception management
- Enhanced productivity of meter readers



M- App & IVRS

Maintenance Survey

- Survey **on-site** using M-App
- Capable of recording 23 types of defects types
- **Photographs mandatory**
- **GPS-based pole location** (Lat-Longitude auto-captured)
- **Pole number selection** from GIS dropdown
- Reports available **division-wise / feeder-wise**

Defect Closure Process

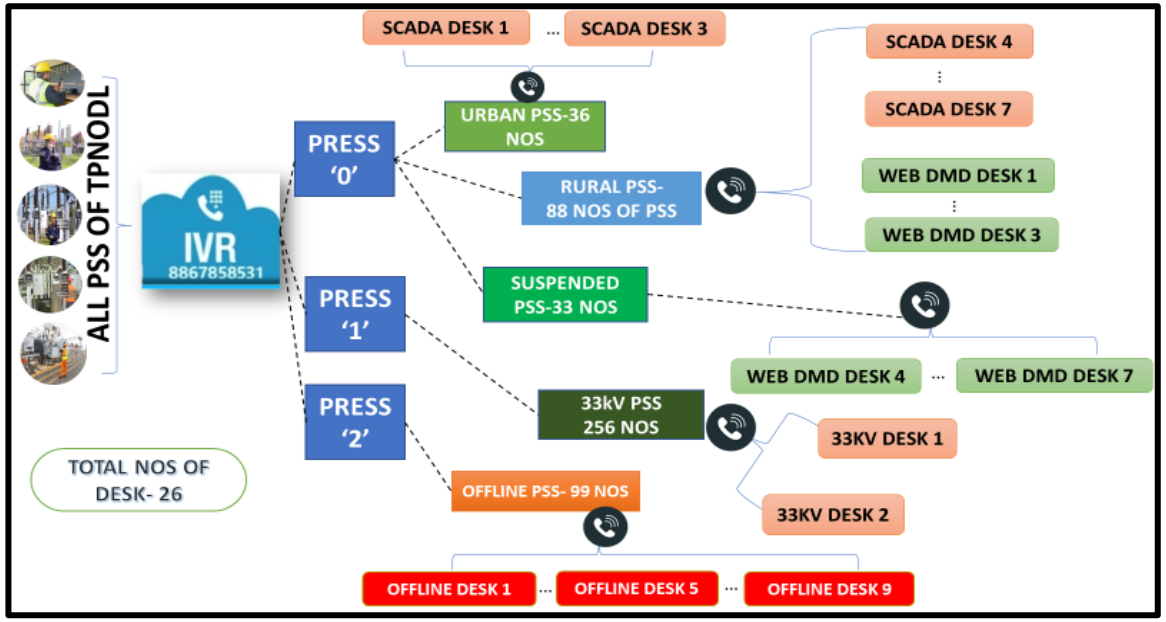
- **Site rectification work under PM Notification**
- Enter **SAP notification number-** Defect status automatically changes to **"Closed"**

➢ Frequency – Quarterly for Urban

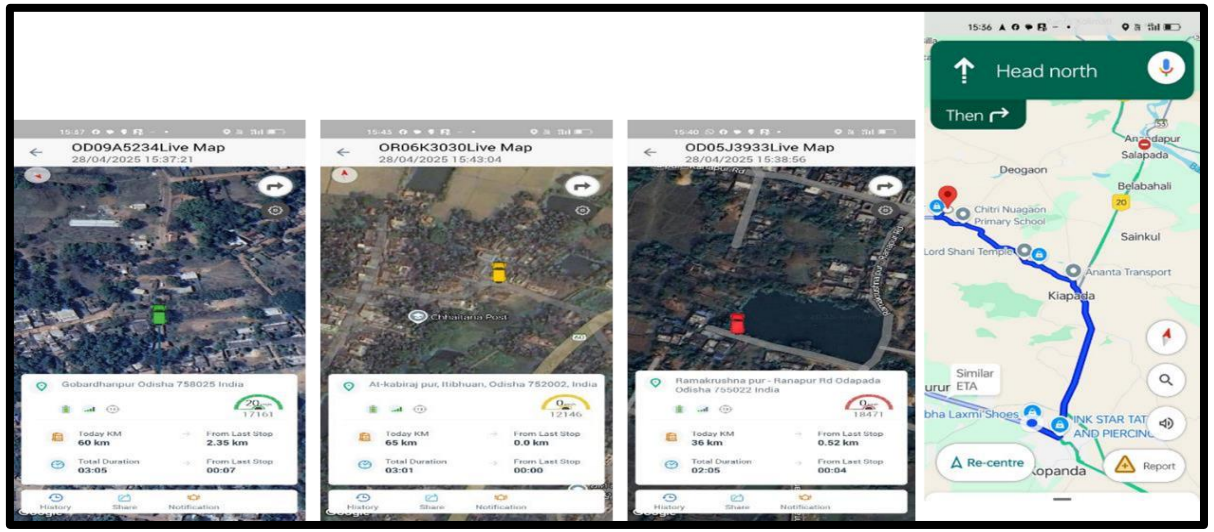
➢ Single platform for **survey, tracking & closure**

IVRS Implementation

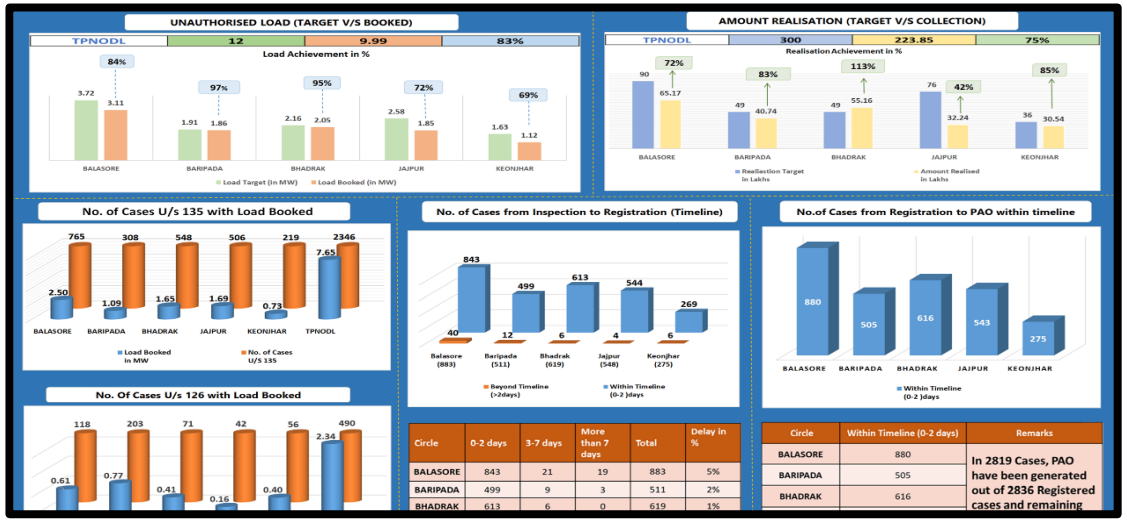
- IVRS for Power System Control Centre Operations – to handle high-volume customer communication channel that routes power issues instantly to the right operational desks.
- Reduces wait times, lowers call abandonment, and delivers faster, data-driven outage response across the utility.



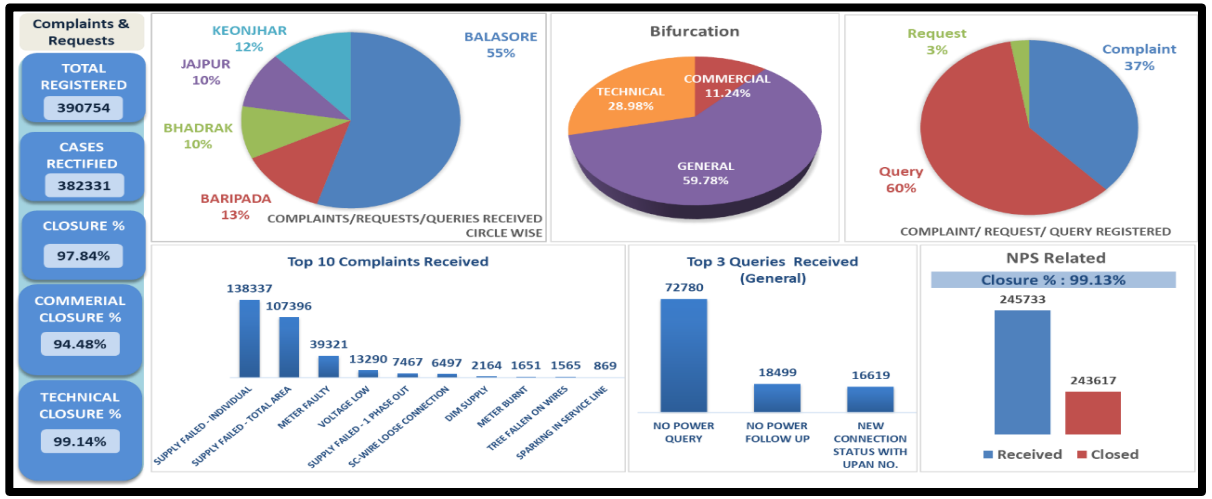
Analytics dashboards & Digitalization Index



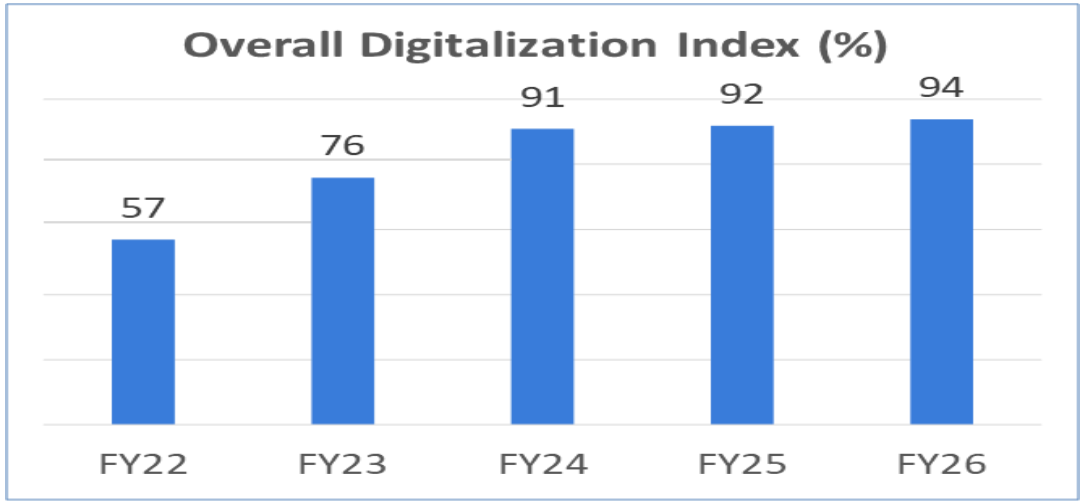
MMG Vehicle Tracking



EAC Dashboard



Complaints & Requests analysis dashboard



Digitalization Index

Way Ahead Plan

SAP PS - CIS - GIS integration for complete project lifecycle tracking and asset synchronization

Integration of SAP PS – CIS – GIS will ensure accurate cost capture, faster project capitalization, timely asset synchronization and financial closure leading to improved cash flow visibility and regulatory reporting accuracy

ADMS integration with GIS

Usage of GIS-provided accurate network model by ADMS for real-time monitoring, control, and optimization of the distribution network

ADMS Integration with Weather and CYME Applications

Improved grid reliability, load forecasting accuracy and accurate contingency analysis

Group consumer portal for smart metered tower connections

Group consumer portal to link all smart metered tower connections associated with a particular service provider. To help monitor consumption pattern, balance and service connection specific information

SK03 App – AI version for PTW Issuance

Centralized way of verifying the site clearance by PSC team like proper isolation. usage of PPE by working staff, usage of proper tools through photo upload at site followed by AI detection

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THANK YOU

For discussions/suggestions/queries email:

sandip.pal@tpnodl.com

Digitalization Index Calculation

Sl No	Department	Process ID	Count of processes	Total Base score	Total achieved score	Digitalisation Index
1	D & IT	10.23.01 to 10.23.23	21	5420	5120	94.00%
2	Operation	10.17.01 to 10.17.16	16	6600	6180	94.00%
3	HR (Human Resource)	10.21.01 to 10.21.18	17	7830	7450	95.00%
4	Finance	10.22.01 to 10.22.16	7	2520	2460	98.00%
5	Business Excellence	10.20.01 to 10.20.10	9	1460	1320	90.00%
6	Commercial	10.18.01 to 10.18.12	9	10372	9641	93.00%
7	Other Shared Service	10.25.01 to 10.25.20	17	2260	920	41.00%
8	NEP (Network Engineering and Project Management)	09.04.01 to 09.04.12	11	3220	3060	95.00%
9	Procurement	10.24.01 to 10.24.04	3	9260	8720	94.00%
10	Customer Service	10.19.01 to 10.19.02	2	970	920	95.00%
		Total Score	112	49912	45791	92.00%